

Game Theory, Mrs Homework

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1.1

Proof. Suppose it is. Then by the definition of properness there is a sequence of totally mixed profiles $\sigma^n \rightarrow (A, V)$ and $\varepsilon_n \rightarrow 0$, $\varepsilon_n > 0$, s.t for each player i and every pair of pure strategies s, s' of that player,

$$u_i(s, \sigma_{-i}^n) < u_i(s', \sigma_{-i}^n) \implies \sigma_i^n(s) \leq \varepsilon_n \sigma_i^n(s'). \quad (*)$$

I write player 1's tremble weights as $(a_n, b_n, c_n, d_n) = (\sigma_1^n(A), \sigma_1^n(B), \sigma_1^n(C), \sigma_1^n(D))$ and player 2's as $(v_n, w_n, x_n, y_n, z_n) = (\sigma_2^n(V), \dots, \sigma_2^n(Z))$. Being totally mixed, all of these are strictly positive; and $\sigma^n \rightarrow (A, V)$ means $a_n, v_n \rightarrow 1$ while the other six weights $\rightarrow 0$.

The idea is to use the two player's constraints against each other. I do it in three steps.

1. Against player 1's trembles, player 2's payoffs from W , X and Y are

$$u_2(W, \sigma_1^n) = a_n + d_n, \quad u_2(X, \sigma_1^n) = a_n + c_n, \quad u_2(Y, \sigma_1^n) = 0,$$

because column Y gives player 2 zero in every row, while W and X each give him a 1 in row A . Since $a_n \rightarrow 1$, for n large both $a_n + d_n$ and $a_n + c_n$ are strictly positive, so Y is strictly worse than W and than X . By $(*)$,

$$y_n \leq \varepsilon_n w_n \quad \text{and} \quad y_n \leq \varepsilon_n x_n.$$

In particular $y_n < w_n$ and $y_n < x_n$ for n large (eventually $\varepsilon_n < 1$).

2. For player 1, against player 2's trembles¹:

$$u_1(B) = v + 2w + 2x + z, \quad u_1(C) = v + w + 2x + y + z, \quad u_1(D) = v + 2w + x + y + z,$$

¹I suppress the superscript n for readability

so, cancelling, the only columns that survive are W, X, Y and

$$u_1(B) - u_1(C) = w_n - y_n, \quad u_1(B) - u_1(D) = x_n - y_n.$$

By Step 1 both differences are strictly positive for n large, so C and D are each strictly worse than B . Applying $(*)$ to player 1,

$$c_n \leq \varepsilon_n b_n, \quad d_n \leq \varepsilon_n b_n, \quad \text{hence} \quad c_n + d_n \leq 2\varepsilon_n b_n < b_n$$

once $\varepsilon_n < \frac{1}{2}$.

3 Finally, step 3:

Now we compare columns V and Z for player 2:

$$u_2(V, \sigma_1^n) = a_n + b_n + c_n + d_n = 1, \quad u_2(Z, \sigma_1^n) = a_n + 2b_n,$$

so

$$u_2(V, \sigma_1^n) - u_2(Z, \sigma_1^n) = (c_n + d_n) - b_n < 0$$

for n large, by 2. Thus V is strictly worse than Z , and $(*)$ gives

$$v_n \leq \varepsilon_n z_n \leq \varepsilon_n \rightarrow 0.$$

But $\sigma^n \rightarrow (A, V)$ forces $v_n \rightarrow 1$. Contradiction. ■

1.2

First, to solve this we need to note two key facts about the matrix of the game:

- (a) column V pays player 1 a 1 in every row (and so does Z — in fact V and Z give player 1 identical payoffs, a constant 1);
- (b) column V pays player 2 a 1 in every row, and no other column ever pays player 2 more than 1, except Z , which pays 2 and only in row B .

I used this above and I will use it here too.

Fix any extensive form Γ with this reduced normal form. I suppose there is an assessment (σ, μ) with $\sigma = (A, V)$ that is sequentially rational and consistent; since Γ is arbitrary, this shows (A, V) is part of a sequential equilibrium in every such tree.

Suppose player 2 plays V , so by (a) player 1 gets 1 no matter which row she ends up playing. Hence at every information set of player 1, every continuation — A included — yields 1, so A is (weakly) optimal there under whatever beliefs she holds. Nothing about Γ entered this; player 1's sequential rationality is automatic in any tree.

By (b), V yields a constant 1 at any information set of player 2, whatever the belief μ . So the only way V could fail to be optimal is if some other column beats 1.

Writing the belief's marginal over player 1's row as $(\mu_A, \mu_B, \mu_C, \mu_D)$,

$$u_2(V) = 1, \quad u_2(W) = \mu_A + \mu_D \leq 1, \quad u_2(X) = \mu_A + \mu_C \leq 1, \quad u_2(Y) = 0, \quad u_2(Z) = \mu_A + 2\mu_B.$$

Only Z can exceed 1, and $u_2(Z) > 1$ exactly when $\mu_B > \mu_C + \mu_D$. So V is a best reply at an information set as

$$\mu_B \leq \mu_C + \mu_D.$$

On path, player 1 plays A with probability 1, so Bayes forces $\mu_A = 1$ (thus $\mu_B = 0$) at every on-path information set of player 2 — and there V is optimal, tying Z at 1. Off path I need beliefs that are limits of Bayes updates from totally mixed trembles. Take any $\sigma^n \rightarrow (A, V)$ in which player 1's tremble onto B is of higher order than onto C and D , say $\sigma_1^n(B) = o(\sigma_1^n(C))$ and $\sigma_1^n(B) = o(\sigma_1^n(D))$. At any information set h of player 2 and node $x \in h$, Bayes gives $\mu^n(x) = P_{\sigma^n}(x)/P_{\sigma^n}(h)$, so the weight of a B -node relative to any A -, C - or D -node in h is a ratio $\sigma_1^n(B)/\sigma_1^n(\cdot)$, which $\rightarrow 0$ (against A because $a_n \rightarrow 1$; against C, D by the choice of orders). Hence in the limit $\mu_B = 0 \leq \mu_C + \mu_D$ at every such h , and V is optimal there.

The one case this misses is an information set of player 2 containing only B -nodes: there $\mu_B = 1$ is forced, and since $u_2(Z) = 2 > 1 = u_2(V)$, V would not be optimal. But such a set cannot arise in a tree with this normal form. If player 2 had a payoff-relevant move reached only after B , he could vary it while holding the rest of his behaviour fixed; the result agrees with V against A, C, D but reaches Z 's outcome against B , i.e. it is a column equal to V in rows A, C, D and to Z in row B — giving player 2 $(1, 2, 1, 1)$. That is none of $V, \dots, Z$, so it would enlarge the reduced normal form. Since the reduced normal form is exactly the five given columns, no such B -only choice exists: every information set of player 2 that contains a B -node also contains an A -, C - or D -node, and the previous paragraph applies. (If some B -only set existed with no payoff-relevant choice, any action — V 's included — is optimal there anyway.)

This argument never used the shape of Γ , so (σ, μ) is a sequential equilibrium with $\sigma = (A, V)$ in every extensive form with this normal form.

Putting the two halves together: (A, V) is sequential in every such tree, yet (Part 1) it is not proper. Hence

$$\text{proper} \subsetneq \bigcap_{\Gamma} \text{sequential eq. of } \Gamma,$$

which is the strict inclusion the notes were pointing at.

2

Setting. We define the russian roulette game in the following game, following what we did in class: we have one player, six chambers, one bullet placed uniformly at random. The player may pull the trigger up to twice. Quitting (at any point) pays 0; being hit pays -8 ; surviving both pulls pays 10.

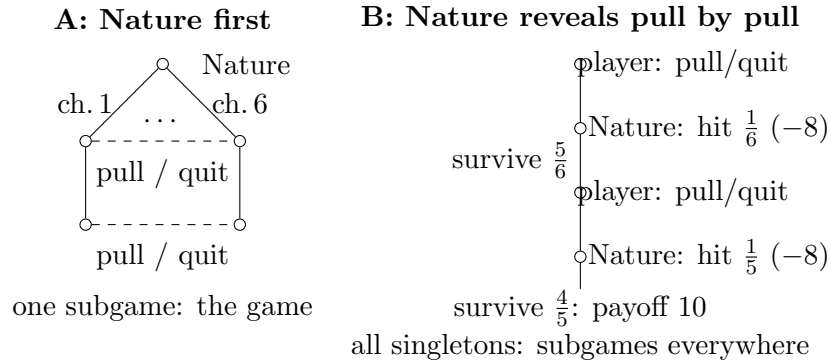
Two representations of the same game. Importantly, there are two natural ways to write the extensive form. This is the key tension of the problem:

- Representation A: Nature moves once, at the start, choosing the loaded chamber (probability $1/6$ each); the player never observes this. Then every decision node of the player

pools the chamber positions he cannot distinguish: his first node is an information set over six positions, and the node after a survived pull is an information set over the five remaining positions. The only proper subgame is the game itself, so subgame perfection has no bite and one needs beliefs.

- Representation B: the player moves first, and Nature resolves the risk pull by pull, publicly: the first pull hits with probability $1/6$; conditional on surviving, the second hits with probability $1/5$ (one bullet, five chambers left). Now every node is a singleton, every continuation is a proper subgame, and the game is solved by backward induction.

In the following figure I represent both cases:



Solution

- **The backward-induction solution (representation B).** At the second node, pulling gives $\frac{4}{5}(10) + \frac{1}{5}(-8) = 8 - 1.6 = 6.4 > 0$, so the player pulls. At the first node, pulling gives $\frac{5}{6}(6.4) + \frac{1}{6}(-8) = \frac{16}{3} - \frac{4}{3} = 4 > 0$, so he pulls. The unique solution is pull–pull, with value 4. Note the conditional probabilities $1/6$ and $1/5$ are part of the tree: at the second node the hazard is a fact, not a belief.
- **PBE is too permissive (representation A).** Now let's try and solve representation A with weak perfect Bayesian equilibrium: beliefs over the chamber at each information set, Bayes' rule on the path, no restriction off the path. I claim the strategy “quit immediately” is a weak PBE, even though backward induction rejects it.

To see why:

Take the strategy (quit at the first node; quit at the second), and the belief, at the second information set, that the bullet sits in the next chamber for sure. The second information set is reached only if the player deviates and pulls, so it is off the path and this belief is allowed. Given the belief, pulling at the second node yields -8 for sure, against 0 from quitting, so quitting there is sequentially rational. Given that continuation, deviating to pull at the first node is worth $\frac{5}{6}(0) + \frac{1}{6}(-8) = -\frac{4}{3} < 0$, so quitting at the start is optimal too. On the path there is nothing for Bayes to check. So (quit, quit) is a weak PBE with value 0 — the player talks himself out of a game worth 4 by adopting, after his own deviation, a belief about the chamber that nothing in his information justifies.

In representation B this profile is simply not subgame perfect: at the second node the

probability is $1/5$ by construction, pulling is worth $6.4 > 0$, and backward induction gives the quit strategy. So the two representations of the same game disagree under (subgame perfection, weak PBE) — weak PBE is strictly too permissive, which is the point of the exercise.

The concept that survives re-representation. What kills the spurious equilibrium is pinning down the belief. The player’s own deviation carries no information about where the bullet is: the chamber was drawn by Nature, independently of anything the player does. So conditional on having survived one pull, the only reasonable belief is uniform over the five remaining chambers — prob $1/5$ — exactly the number representation B hardwires into the tree. Sequential equilibrium delivers precisely this: consistency requires beliefs to be limits of Bayes updates under totally mixed strategies, and since the player’s trembles are independent of Nature’s draw, the limit belief at the second information set must be the uniform conditional, whatever the equilibrium strategy is. With that belief the second pull is worth 6.4, the first is worth 4, and the unique sequential equilibrium of representation A is pull-pull — the backward-induction solution of representation B. For this class of games (beliefs are about Nature only), the same argument applies in any representation, so sequential equilibrium is the concept whose strategy prediction is representation-independent.

So...which finite games with Nature admit such a representation. The construction that turns A into B is: delay Nature’s move, split it into conditional pieces, and reveal each piece publicly at the moment it first matters (here: chamber drawn once, unobserved, becomes hit-or-miss taken each time with probs $1/6, 1/5, \dots$). This works whenever two conditions hold:

First, no player is privately informed about Nature: if some player observed the draw and another did not (types), the uninformed player would have to move without knowing it in every representation, and a nontrivial information set survives.

Second, the personal players themselves have perfect information (no simultaneous moves among them), since an information set created by another player’s unobserved move cannot be removed by re-timing Nature.

Under these two conditions — all uncertainty is common uncertainty about Nature, and players otherwise see everything — Nature’s draws can be postponed and resolved in public step by step, every information set becomes a singleton, and subgame perfection (backward induction) is the natural solution concept.

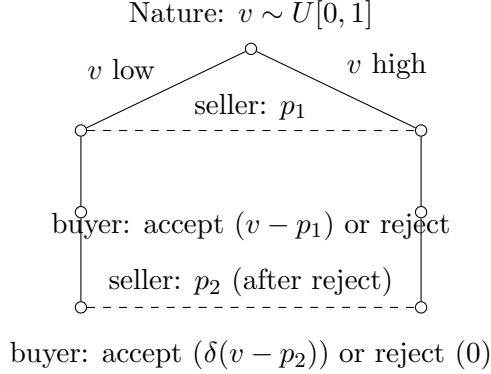
3

Setting. I use the seller-offer game with two periods and $\delta = \frac{1}{2}$. The seller’s cost is 0. In period 1 the seller sets p_1 ; buyers accept or reject; in period 2 the seller sets p_2 to whoever has not bought. A value- w agent who buys at price p in period t gets $\delta^{t-1}(w - p)$, and 0 if it never buys; the seller collects the discounted prices paid. We allow two versions

1. one buyer, value $v \sim U[0, 1]$, privately known.
2. a unit mass of buyers $i \in [0, 1]$, buyer i has value i , common knowledge

A buyer-side strategy is the same object in both — a map $\beta(w, h)$ giving the probability that the agent of value w accepts at history h , read as “what type w does” in (i) and “what buyer $i = w$ does” in (ii) — and the seller’s strategy is a price after each history. So it makes sense to ask whether the two solution concepts select the same pairs (β, π) .

The tree of version (i) looks as follows (two representative values drawn; the dashed line is the seller’s information set, the reason beliefs are needed). Version (ii) is the same picture with Nature’s move deleted: each value is a separate buyer at her own node, so there is no dashed line and every continuation is a subgame.



Solution/prediction First solve version (ii) by backward induction. If the leftover population is $[0, \kappa)$, the seller’s last-period revenue from p_2 is $p_2(\kappa - p_2)$, maximised at $p_2 = \kappa/2$. Anticipating this, the period-1 cutoff type is indifferent: $\kappa - p_1 = \delta(\kappa - \kappa/2) = \kappa/4$, so $\kappa = \frac{4}{3}p_1$. The seller then chooses p_1 to maximise $p_1(1 - \frac{4}{3}p_1) + \frac{1}{2} \cdot \frac{2p_1}{3} \cdot \frac{2p_1}{3} = p_1 - \frac{10}{9}p_1^2$, giving

$$p_1 = 0.45, \quad \kappa = 0.6, \quad p_2 = 0.3, \quad \text{revenue } 0.225.$$

Since the residual after any p_1 — including off-path prices — is the determinate set $[0, \frac{4}{3}p_1)$ and the continuation is pinned at every node, this is the unique SPE outcome.

In version (i), on the equilibrium path, the same numbers come out and it is interesting seeing why. After $p_1 = 0.45$ is rejected, Bayes rule gives the seller the posterior $U[0, 0.6)$, and his problem is $\max_{p_2} p_2 \frac{0.6 - p_2}{0.6}$, which is the continuum’s problem $\max_{p_2} p_2(0.6 - p_2)$ divided by the constant 0.6: integrating over a uniform type is averaging over a uniform population, so the maximiser is the same $p_2 = 0.3$. The buyer side needs no assumption at all, and this is the genuinely “Harsanyi part”: a single agent cannot move the leftover set — in (ii) because she has measure zero, in (i) because the seller responds to the signal “reject” through the strategy β , so a lone deviating type just pools with the equilibrium rejectors — hence every value faces the identical problem “take $v - p_1$ now or $\delta(v - p_2)$ tomorrow” in both games.

Off-Path. So the two concepts can only disagree off the path, and this is the substance of the problem as posed.

Consider the following profile in version (i): the seller sets $p_1 = \frac{1}{2}$; after the on-path rejection, Bayes gives $U[0, \frac{2}{3})$ and the seller sets $p_2 = \frac{1}{3}$ (types accept by the cutoff rule; the indifferent type is $\frac{2}{3} - \frac{1}{2} = \frac{1}{2}(\frac{2}{3} - \frac{1}{3})$, which checks). Off the path: if the seller ever sets $p'_1 \neq \frac{1}{2}$, let his belief after the subsequent rejection be the point mass at $v = 0$, and let set quote $p_2 = 0$.

This is sequentially rational: against the belief $v = 0$ every price earns zero, so setting 0 is (weakly) optimal. Buyers best respond to the actual continuation: anticipating a price of 0 tomorrow, a value v accepts p'_1 only if $v - p'_1 \geq \delta v$, i.e. $v \geq 2p'_1$, so a deviation earns the seller at most $\max_{p'_1} p'_1(1 - 2p'_1) = \frac{1}{8}$. The path earns $\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{9} \approx 0.222 > \frac{1}{8}$, so no deviation is profitable and this is a weak PBE. (Nothing special about $\frac{1}{2}$: the same “surrender” threat supports a whole interval of first-period prices.)

The identified profile is not an SPE of version (ii). There, after the deviation p'_1 , the leftover population is the fact $[0, 2p'_1)$, and against it $p_2 = 0$ earns zero while $p_2 = p'_1$ earns $p_1'^2 > 0$: the surrender threat is not credible when the seller knows who is still in the market. The whole trick in (i) was the belief — the seller “threatens himself” by adopting, after his own deviation, a posterior ($v = 0$) that his information does not justify: the buyer has done nothing new, so there is nothing to update on. The gap between the two equilibrium sets is exactly the set of profiles produced by such beliefs.

Proposed solution The fix is therefore a single restriction on the beliefs of version (i):

(B) at every history, on or off the path, the seller’s posterior is the Bayes update of the prior given the buyer’s strategy β ; in particular it does not move in response to the seller’s own prices.

This is the “no signalling what you don’t know” condition of Fudenberg-Tirole: v is independent of the seller’s actions, so only the buyer’s accept/reject record is informative. Since a rejection is consistent with all low values, the conditioning event always has positive probability and (B) pins the posterior uniquely — it is the truncated uniform $U[0, \theta]$, i.e. precisely the residual population of version (ii) at the same history. Under (B) the surrender belief is illegal, the seller’s problem at every node — not just on the path — coincides with the continuum’s up to a positive constant, and the buyer side coincided already; hence the equilibrium strategy pairs of the two versions are the same, and the example above shows (B) cannot be dropped.